

These resources are abundant and have no such side effects. But to get the maximum benefit, we need to maximise the efficiency of conversion and find new methods while dealing with the challenges that might arise.

For instance, a group of scholars has come up with a way to store solar energy so as to use it when sunlight is not available. The phenomenon is called photoswitching, which allows molecules known as photoswitches to absorb solar energy and transfer the energy between its configurations for a long period of time. This article discusses some of the lesser-known ways of harnessing electricity.

Snow

In numerous places worldwide, snowing is a common phenomenon in winters. In such situations, the accumulation of snow decreases the amount of sunlight that reaches the solar array. This makes solar panels an ineffective option for supplying power.

In 2019, the University of California (UCLA) researchers and colleagues designed a device—a snow-based triboelectric nanogenerator or snow TENG—that creates electricity from falling snow. It makes use of the fact that snow is positively charged and gives up electrons.

When falling snow contacts the

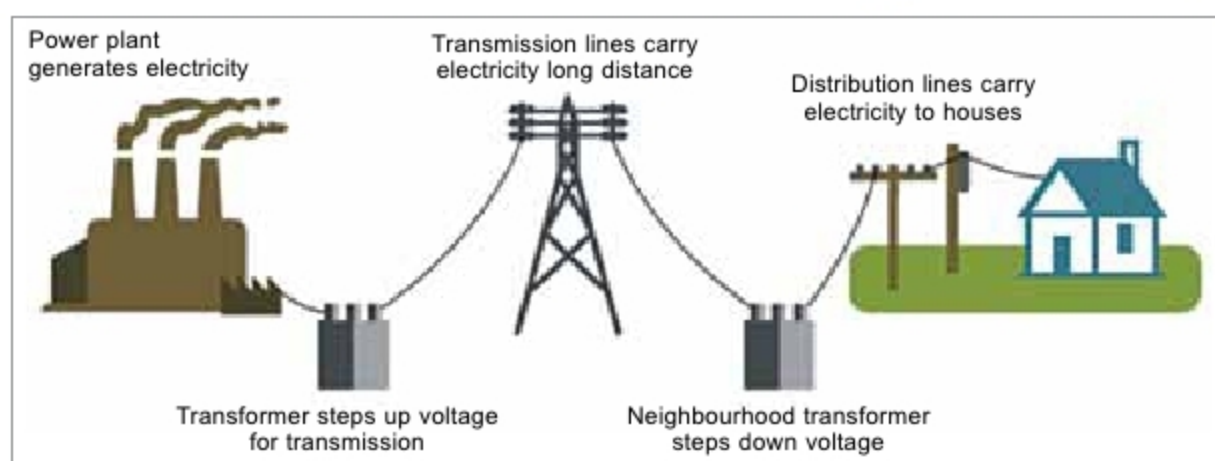


Fig. 1: Conventional electric grid (Credit: www.researchgate.net)

surface of a negatively charged material, silicone layer in this case, it produces a charge that the electrode in the device captures, thereby creating electricity. Silicone is chosen because it easily extracts electrons when compared to other materials such as Teflon and aluminium foils. It also simplifies the fabrication process since silicone is easily available, making the device affordable.

This device made using 3D printing can ensure an uninterrupted power supply on integration with solar panels even in winter months. It does not require batteries for generating electricity and hence is suitable for even remote areas. The unique information provided by this device includes the amount of snow falling at a time, the direction of snowfall, velocity of wind, and much more.

Bacteria

While several bacteria are harmful to humans and give rise to a host of diseases, some others are useful. Illnesses from pathogenic bacteria include staph infections, cholera, strep throat, tuberculosis, and many more. Researchers in the US have discovered that harnessing a combination of phototropic (light-consuming) and heterotrophic (matter-consuming) bacteria in microbial fuel reactions could generate currents that are seventy times more powerful than in usual setups.

For this, the microbes are connected to cathode and anode in a microbial fuel cell, so that transfer of electrons due to oxidation occurs, which generates power.

In another instance, researchers have developed a device known as an air-gen or air-powered generator that employs a bacteria-derived protein to produce electricity from moisture present in the atmospheric air. The microorganism *Geobacter* is used for these ultrasmall (less than ten micrometres thick) electrically conductive protein wires. This is a non-polluting and economical way of producing renewable energy out of mere thin air 24/7, even in indoor environments, unlike most other renewable energy sources. Also, it is possible to generate electricity in places that have extremely low humidity, such as deserts, using this method.

Plastic

Plastics and their byproducts are quite hazardous for humans and the environment as these are non-biodegradable and lead to a host of health problems. A minute percentage of plastics get recycled every year.

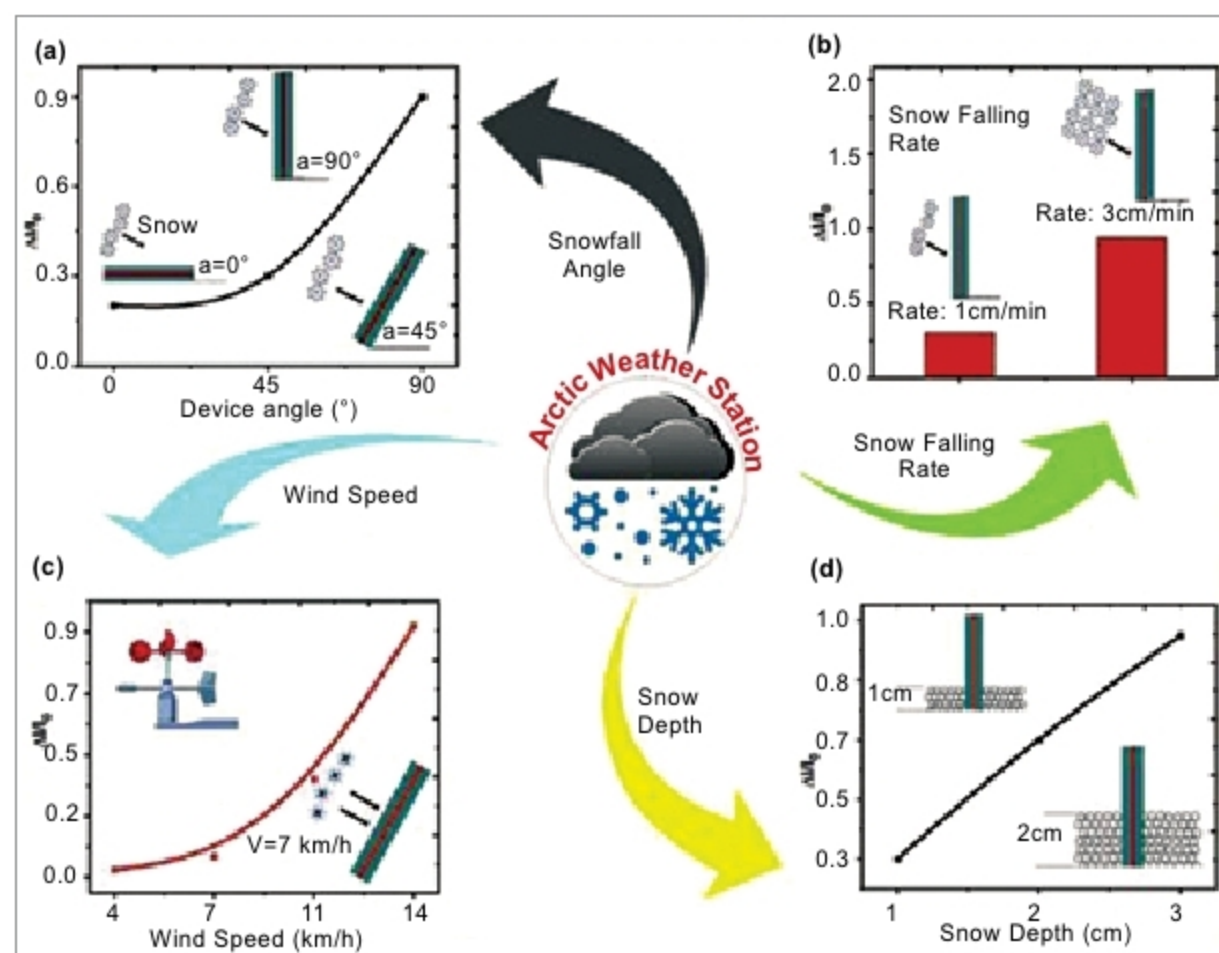


Fig. 2: Data available from snow TENG (Credit: www.sciencedirect.com)